Code & Data

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National Institute of Mental Health

Who Funds Open Science? Tracking Data Sharing in Publications Across Major Funders

Dashboard

opensciencemetrics.org

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INTRODUCTION

Open data sharing is central to the reproducibility and transparency of biomedical research, anchored by the FAIR principles for data stewardship (*1*). Funders, journals, and institutions increasingly *mandate* sharing, yet measuring compliance at population scale remains difficult.

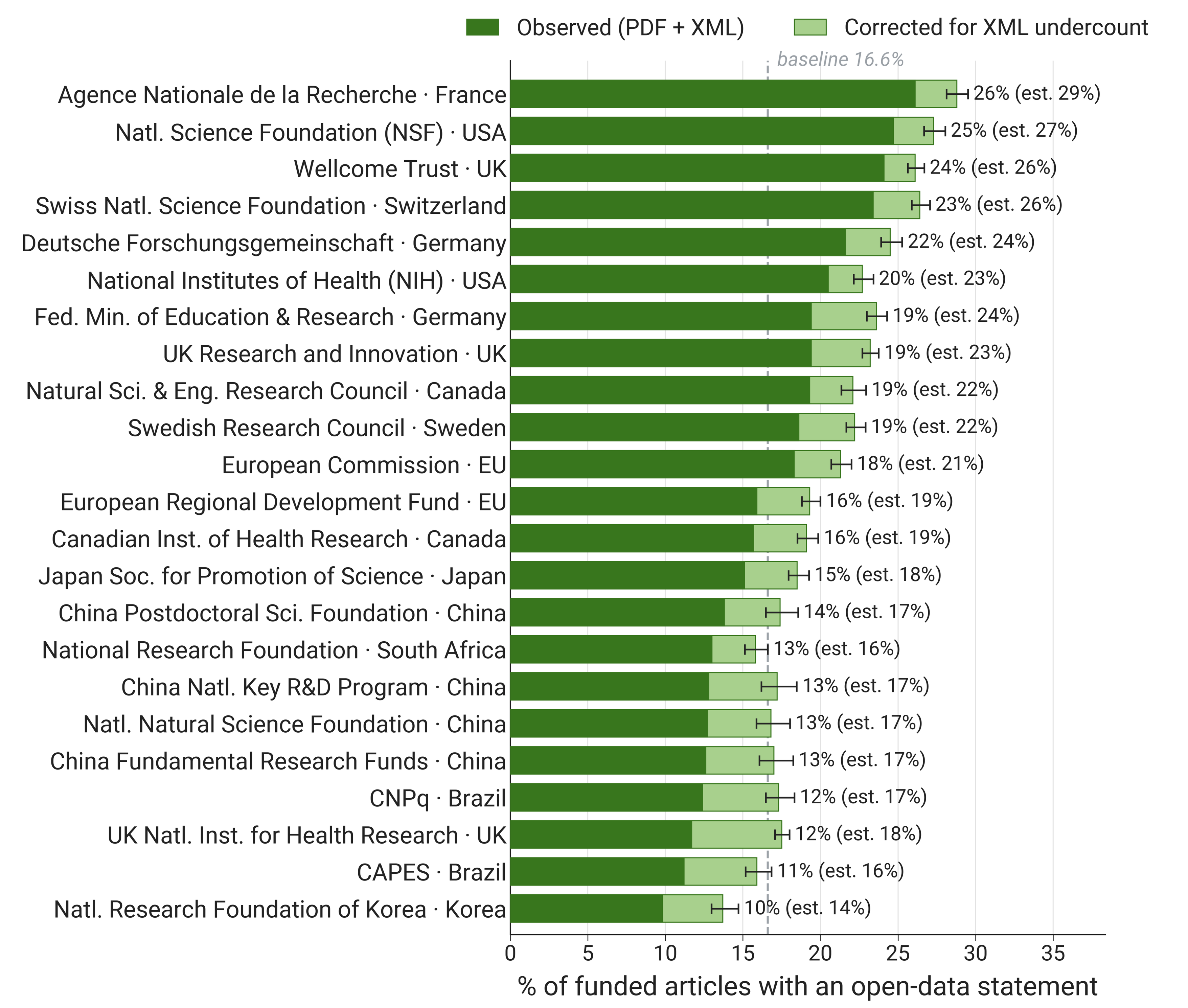
Text mining XML versions of open-access articles systematically *undercounts* data sharing because availability statements are often added late in production or appear in sections omitted from XML. In a landmark XML-based survey, open data rates reached only ∼15% by 2020 (*2*).

We measured open data sharing across ∼784,000 recent biomedical articles using a **PDF-first** detection pipeline and benchmarked rates across major funders and journals.

METHODS

- 784,262 open access PubMed Central articles, January 2024 – June 2025.
- Full text from 256,660 PDFs (32.7%) via **MinerU** (*3*); remaining 527,602 (67.3%) from PMC XML.
- Open data & open code statements detected with **oddpub v7.2.3** (*4*) via curated pattern matching.
- Funder, journal, and institution metadata from **OpenAlex** (*5*); 816 funders normalized (e.g. NIH institutes aggregated to parent).
- **Major-funder filter (figure).** To compare peer agencies, the funders figure is restricted to funders with ≥100,000 total works indexed in OpenAlex *and* above a Weibull-derived size threshold (≥2,586 funded articles); the full ranking of all 816 funders is reported in the preprint.
- Observed rates are census-exact corpus proportions (no sampling interval). Journal-level **correction factors** from head-to-head PDF vs. XML comparison *impute* the open-data statements PDF processing would have found on XML-only articles; the corrected estimate carries a 95% Wilson-score **imputation interval**.
- Cross-sectional, descriptive design: reported differences are *associations*, not causal policy effects.
- Recent mandates for data availability statements can lower the sensitivity of detection tools (*6*).
- All results explorable via an interactive dashboard at `opensciencemetrics.org`.

RESULTS: FUNDERS



Major biomedical funders, ranked by open-data rate (see *Methods* for the size filter). Solid bars: observed rate (PDF + XML); lighter bars: corrected estimate — because mining XML alone misses data-availability statements present in the PDF, XML-only articles are adjusted upward using PDF-vs-XML correction factors; whiskers: 95% imputation intervals on the corrected estimate (observed bars are census-exact, no interval); dashed line: 16.6% funded-article baseline. Observed rates ranged from ∼10% to ∼26%: agencies with strong open-science cultures (ANR, the US and Swiss science foundations, Wellcome, DFG) reached up to ∼**1.6**× the baseline, while the highest-volume funders (NSFC China, NIH) sat closer to it. Smaller and regional funders are excluded here, but several would rank at the very top — e.g. the *National Science Foundation of Sri Lanka* (27.7%) and data-intensive research centers such as EMBL and SciLifeLab (45–53%). The preprint ranks all 816 funders in full.

REFERENCES

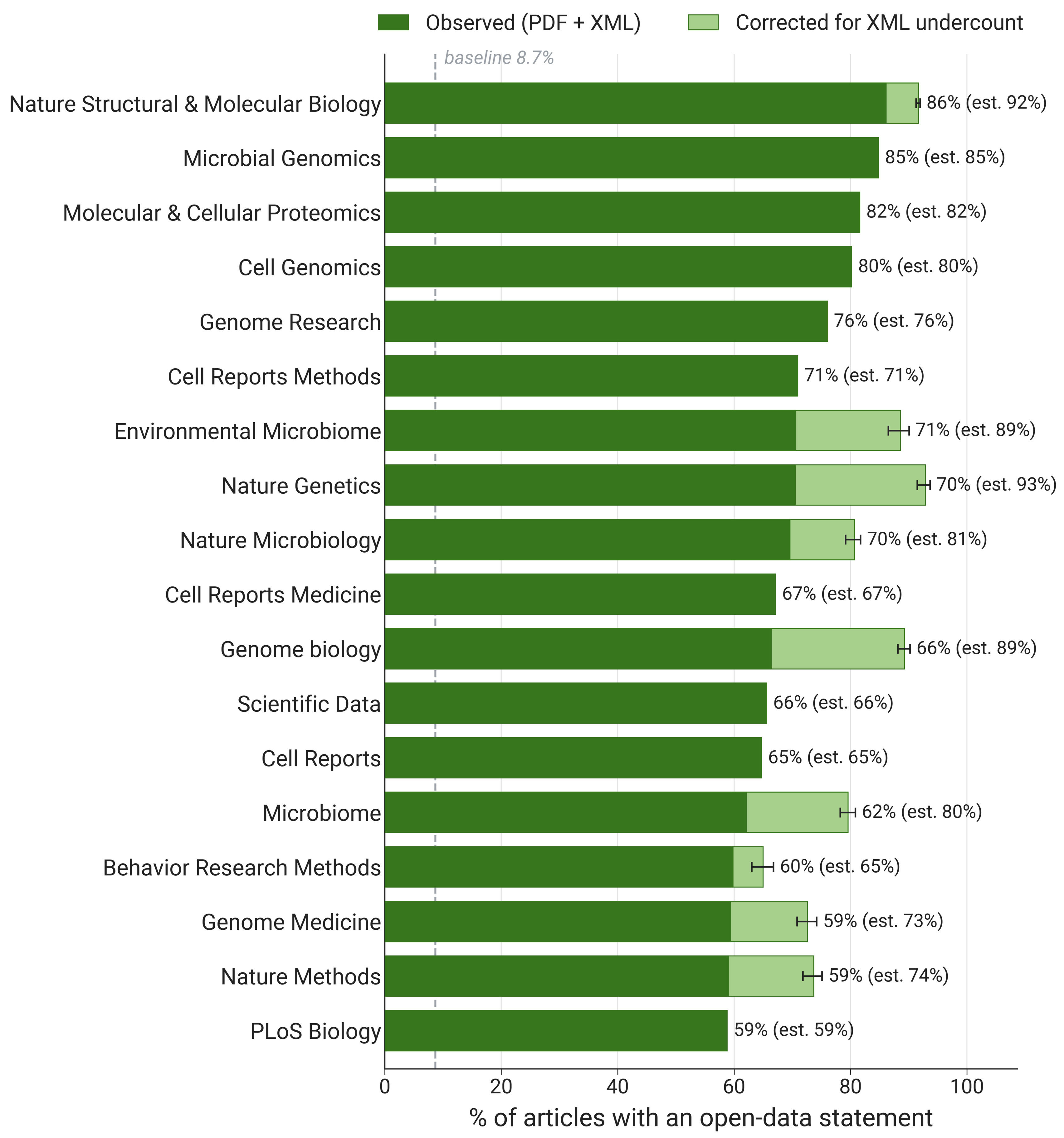
1. M. D. Wilkinson *et al.*, *Sci. Data* **3**, 160018, DOI 10.1038/sdata.2016.18 (Mar. 2016).
2. S. Serghiou *et al.*, *PLoS Biol.* **19**, e3001107, DOI 10.1371/journal.pbio.3001107 (Mar. 2021).
3. B. Wang *et al.*, *arXiv preprint arXiv:2409.18839*, DOI 10.48550/arXiv.2409.18839 (2024).
4. N. Riedel *et al.*, *Data Sci. J.* **19**, 42, DOI 10.5334/dsj-2020-042 (2020).
5. J. Priem *et al.*, *arXiv preprint arXiv:2205.01833*, DOI 10.48550/arXiv.2205.01833 (2022).
6. A. Iarkaeva *et al.*, *PLoS One* **19**, e0302787, DOI 10.1371/journal.pone.0302787 (May 2024).
7. H. Deeb *et al.*, *PLoS One* **20**, e0328065, DOI 10.1371/journal.pone.0328065 (July 2025).

TAKEAWAYS

- Open-data rates varied **more than tenfold** across funders and journals.
- Open-data rate **8.7%** across all articles, but **16.6%** (∼2-fold) among articles that report funding.
- PDF-based detection found ∼**52% more** data-sharing statements than XML alone.
- Major funders reach observed rates above 20% (corrected up to ∼28%); top journals reach 70–86% observed, with corrected estimates up to **92.9%** (Nature Genetics).
- Data sharing rates vary *consistently* by funder, helping funders identify which policies and practices most effectively incentivize sharing.
- **New result from ICSSI hackathon!** Cross-checking our funder identification against Dimensions' fuller, publisher-sourced links, the ranking is largely robust (Spearman 0.88) — though the *European Commission* appears under-credited, rising to #2 under more complete attribution.

Scan to see Dimensions figure

RESULTS: JOURNALS



The 18 highest-rate journals (≥150 articles). Bars, whiskers, and shading as in the funders figure; lighter bars use journal-level correction factors from head-to-head PDF vs. XML comparison; dashed line: 8.7% overall baseline. Across all 1,389 journals, rates spanned below 5% to above 90%. The leaders shown here are dominated by genomics and structural-biology titles, with psychology methods journals (e.g. *Behavior Research Methods*) also prominent — reflecting post-replication-crisis norms.

DISCUSSION

- Variation clusters by **disciplinary norms** (genomics and psychology journals lead) and by funder/journal policy **stringency and enforcement** — not the mere existence of a mandate.
- Detecting a *statement* is not the same as verifying *accessible* data; reported rates are upper bounds. External validation suggests current rates are conservative (*7*).
- Because funders and journals are not randomly assigned to articles, differences reflect a confounded mix of policy, discipline, infrastructure, and self-selection.
- **Future work:** expand PDF coverage and extend the window to enable longitudinal analysis of policy changes (e.g. the 2023 NIH Data Management & Sharing Policy).

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Generative AI. The authors used generative AI tools (Anthropic's Claude, OpenAI models, the Cursor editor) as drafting, editing, and coding assistants. All numerical results were verified against the underlying analysis outputs. Generative AI was not used to fabricate, alter, or substitute for the underlying data.

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